HEARME: AN AI-POWERED INCLUSIVE E-LEARNING PLATFORM FOR HEARING-IMPAIRED STUDENTS

Group Research Dissertation

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DECLARATION

We declare that this our own work & this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning & to the best of our knowledge & belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

HearMe is a comprehensive, AI-powered inclusive e-learning platform designed to address the persistent educational barriers faced by hearing-impaired students in Sri Lanka. The platform integrates four interconnected modules: an Interactive Sri Lankan Sign Language (SSL) Self-Learning and Gesture Evaluation Module, an AI-Powered Sinhala Lip-Reading Assistant with 3D Avatar Feedback, a Teacher-Student Engagement Module with Emotion-Adaptive Learning, and a Multi-Sensory Music Experience Platform (Resonance). Together, these modules provide a holistic digital learning ecosystem that simultaneously addresses sign language acquisition, pronunciation learning, emotional engagement monitoring, and music accessibility.

The SSL module employs MediaPipe Holistic for real-time landmark extraction, a hybrid LSTM and CNN deep learning pipeline for gesture recognition, and a dual-format sign demonstration system combining reference video with three-dimensional avatar rendering, achieving classification accuracy between 88.89% and 100% across vocabulary categories. The lip-reading assistant applies a CNN-LSTM temporal sequence model on a custom Sinhala-specific lip-movement dataset, with phoneme-to-viseme mapping and a 3D avatar feedback engine to guide hearing-impaired learners through visual speech practice. The emotion-adaptive engagement module integrates the DeepFace facial expression recognition library, achieving approximately 97% face detection accuracy and 80% emotion classification accuracy, coupled with an intelligent adaptive intervention engine that triggers Sign Language quizzes and visual exercises when disengagement is detected. The Resonance music platform combines a CNN trained on mel-spectrogram representations of the DEAM dataset for real-time music emotion recognition, a custom ESP32-based haptic feedback device communicated through the Web Serial API, Google Cloud Speech-to-Text transcription supporting eleven languages, and an AI-powered contextual music guide powered by Groq Llama 3.1.

The system is implemented as a full-stack web platform with a React frontend, Flask and FastAPI microservice backends, TensorFlow/Keras model inference layers, and a MERN stack teacher dashboard. Evaluation through functional testing, user acceptance testing with hearing-impaired participants, and performance benchmarking demonstrates strong accuracy, usability, and inclusivity across all four modules. The platform advances the state of accessible educational technology in Sri Lanka and provides a replicable, modular architecture for future development of inclusive e-learning systems.

Keywords: Sri Lankan Sign Language, gesture recognition, LSTM, CNN, lip-reading, emotion detection, adaptive learning, music accessibility, haptic feedback, inclusive e-learning, hearing impairment

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
BLE	Bluetooth Low Energy
CNN	Convolutional Neural Network
CORS	Cross-Origin Resource Sharing
DEAM	Database for Emotional Analysis in Music
DeepFace	Deep Learning-based Facial Analysis Library
ESP32	Espressif Systems 32-bit Microcontroller
FER	Facial Expression Recognition
FFT	Fast Fourier Transform
GLB	Binary glTF 3D Model Format
GUI	Graphical User Interface
HCI	Human-Computer Interaction
HOG	Histogram of Oriented Gradients
IT	Information Technology
LLM	Large Language Model
LRA	Linear Resonant Actuator
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MERN	MongoDB, Express.js, React.js, Node.js
MER	Music Emotion Recognition
MIR	Music Information Retrieval
ML	Machine Learning
MSE	Mean Squared Error
NLP	Natural Language Processing
PWM	Pulse Width Modulation
REST	Representational State Transfer
ROI	Region of Interest
SLIIT	Sri Lanka Institute of Information Technology
SLR	Sign Language Recognition
SSL	Sri Lankan Sign Language
STFT	Short-Time Fourier Transform

Abbreviation	**Description**
TTS	Text-to-Speech
UAT	User Acceptance Testing
UDL	Universal Design for Learning
UI	User Interface
UX	User Experience
VSR	Visual Speech Recognition
WebRTC	Web Real-Time Communication
WHO	World Health Organization

INTRODUCTION

Education is widely recognised as one of the most fundamental rights afforded to every individual, irrespective of physical or sensory ability. However, for the estimated 430 to 466 million people worldwide who experience some form of disabling hearing loss (World Health Organization, 2021), access to mainstream educational systems remains fraught with significant and persistent barriers. In the context of Sri Lanka, approximately 1.6% to 9% of the population equating to hundreds of thousands of individuals live with hearing-related disabilities, many of whom are excluded from conventional pedagogical frameworks that rely heavily on auditory communication.

The proliferation of e-learning platforms in the post-COVID-19 era has transformed how education is delivered globally. Yet this digital revolution has largely overlooked the specialised requirements of hearing-impaired learners. Existing online learning platforms have made commendable strides in general accessibility, but they lack sophisticated mechanisms for real-time emotional engagement monitoring, sign language learning with automated feedback, visual pronunciation training, and multi-sensory music experience delivery tailored to the non-hearing population. Hearing-impaired students were disproportionately disadvantaged by the rapid shift to remote learning, as standard online education formats rely heavily on audio-based instruction, verbal communication, and multimedia content without adequate captioning or sign language support.

HearMe is a comprehensive, AI-powered inclusive e-learning platform developed to bridge this gap, specifically targeting hearing-impaired students in Sri Lanka and the broader South Asian context. The platform integrates four purpose-built, interconnected modules: an Interactive Sri Lankan Sign Language (SSL) Self-Learning and Gesture Evaluation Module, an AI-Powered Sinhala Lip-Reading Assistant with 3D Avatar Feedback, a Teacher-Student Engagement Module with Emotion-Adaptive Learning, and a Multi-Sensory Music Experience Platform named

Resonance. Together, these modules address the full spectrum of educational accessibility needs identified through review of the existing literature and through direct engagement with hearing-impaired learners in Sri Lanka.

1.1 Background and Literature Review

1.1.1 Hearing Impairment and Educational Exclusion in Sri Lanka

Hearing impairment encompasses a wide spectrum of conditions ranging from partial hearing loss to complete deafness, significantly affecting an individual's ability to communicate and access education. In Sri Lanka, it is estimated that approximately 9% of the population experiences some form of hearing loss, with a considerable proportion classified as profoundly deaf [1]. Globally, the number of individuals affected by hearing loss is expected to increase dramatically, reaching nearly double the current figures by 2050, indicating a growing need for inclusive and accessible educational solutions [2].

Sinhala Sign Language (SSL) serves as the primary communication medium for the Deaf community in Sri Lanka. Unlike globally recognized systems such as American Sign Language and British Sign Language, SSL is a unique linguistic system with its own grammar and structure. However, most existing digital learning platforms fail to recognize SSL as an independent language, resulting in limited availability of localized educational tools.

The transition to digital learning environments during the COVID-19 Pandemic further exposed these limitations. Studies conducted within Sri Lankan universities highlight that hearing-impaired students faced significant challenges in adapting to online learning due to lack of accessibility features such as sign language interpretation, visual learning aids, and interactive engagement mechanisms [3]. Similar findings were observed in international contexts, where Deaf students reported difficulties in understanding content, participating in discussions, and maintaining engagement in virtual classrooms [4].

Beyond communication barriers, hearing-impaired learners encounter additional challenges in areas such as speech articulation, emotional engagement, and sensory-based learning. Traditional educational systems often rely heavily on auditory instruction, making it difficult for Deaf learners to fully participate. Moreover, the absence of tools that support visual speech recognition, emotional awareness, and alternative sensory experiences further contributes to educational exclusion.

1.1.2 Prior Work in Sign Language Learning Technology

Early developments in Sri Lankan sign language technology primarily focused on translation systems rather than learning-oriented platforms. Systems such as EasyTalk introduced machine learning techniques to translate spoken language into Sinhala Sign Language using gesture recognition and image classification [5]. Similarly, UTalk utilized image processing and machine learning to convert sign language gestures into text and speech outputs [6]. While these systems demonstrated the feasibility of automated sign language recognition, they were limited in their ability to support structured learning or provide real-time feedback to users.

Subsequent research expanded into e-learning platforms designed specifically for hearing-impaired students. These systems enabled content delivery through visual materials and teacher-uploaded resources, facilitating asynchronous learning [7], [8]. However, such platforms often lacked mechanisms for interactive assessment, gesture accuracy evaluation, and personalized learning support.

A notable advancement in this domain is the Hastha platform, which integrates machine learning techniques such as MediaPipe Holistic and Long Short-Term Memory (LSTM) networks for sign language recognition and evaluation [9]. Hastha demonstrated the effectiveness of landmark-based sequence modeling for Sinhala Sign Language and provided a foundation for gesture-based learning systems. However, its capabilities remain limited in several aspects, including the absence of multi-modal instructional methods, lack of detailed feedback mechanisms for complex gestures, and limited adaptability to individual learner progress.

Advancements in deep learning have significantly contributed to gesture recognition technologies. Long Short-Term Memory networks are particularly effective in handling sequential data, making them suitable for modeling temporal patterns in sign language gestures. Frameworks such as MediaPipe Holistic enable real-time extraction of body, hand, and facial landmarks, facilitating accurate gesture representation. Additionally, datasets such as SSL400 provide a standardized collection of Sinhala Sign Language gestures, enabling the development and evaluation of machine learning models tailored to the Sri Lankan context [12].

Despite these advancements, existing systems primarily focus on isolated functionalities and do not offer a comprehensive learning environment that integrates gesture recognition, interactive feedback, and user-centered design.

1.1.3 Visual Speech Recognition and Lip-Reading Research

While sign language serves as a primary communication method, speech development and lip-reading remain essential skills for many hearing-impaired individuals. In Sri Lanka, there is a notable lack of tools that support visual speech recognition tailored to the Sinhala language, creating a gap between sign language comprehension and spoken communication.

Research in visual speech recognition has demonstrated the effectiveness of combining convolutional neural networks (CNNs) with temporal modeling techniques to capture both spatial and dynamic aspects of speech. Lip-reading systems rely on analyzing sequences of mouth movements, requiring models capable of understanding temporal dependencies. This makes architectures such as CNN-LSTM combinations particularly suitable for such tasks.

Furthermore, the use of three-dimensional avatars has emerged as a powerful instructional tool in assistive learning environments. Unlike traditional video-based demonstrations, 3D avatars allow users to observe articulatory movements from multiple angles, adjust playback speed, and focus on specific aspects of speech production. This enhances the learner's ability to understand and replicate complex mouth movements, thereby improving speech articulation skills.

However, existing sign language learning platforms do not effectively integrate lip-reading assistance with gesture-based learning, resulting in fragmented learning experiences for users.

1.1.4 Emotion-Aware Educational Systems

The integration of emotional intelligence into educational systems has gained increasing attention in recent years. The concept of affective computing, introduced by Rosalind Picard, emphasizes the importance of recognizing and responding to human emotions in computational systems. In educational contexts, emotional states such as engagement, confusion, and frustration have been shown to significantly influence learning outcomes.

Recent advancements in computer vision and machine learning have enabled the development of systems capable of detecting emotional states through facial expression analysis. Techniques utilizing convolutional neural networks and pre-trained models have achieved high accuracy in recognizing emotions under controlled conditions [10]. Libraries such as OpenCV and MediaPipe further facilitate real-time facial landmark detection, making emotion-aware systems more accessible for web-based applications.

In online learning environments, monitoring student engagement is particularly important. Studies have demonstrated that systems capable of identifying disengagement or confusion can adapt instructional strategies in real time, improving overall learning effectiveness. However, existing platforms for hearing-impaired learners, including those focused on sign language, do not incorporate emotion-aware features, limiting their ability to provide personalized and adaptive learning experiences.

1.1.5 Music Accessibility and Multi-Sensory Learning

Music and sensory-based learning represent an underexplored area in assistive education for hearing-impaired individuals. Although traditionally considered inaccessible to Deaf learners, research has shown that music can be experienced through alternative sensory channels such as vibration and visual representation.

Studies indicate that hearing-impaired individuals can perceive rhythm and emotional content in music through vibrotactile feedback, where sound vibrations are transmitted through physical surfaces. This suggests that multi-sensory learning approaches can enhance engagement and provide new pathways for understanding abstract concepts.

Music Emotion Recognition (MER) further contributes to this domain by enabling systems to interpret and represent the emotional characteristics of music using computational models. Frameworks such as Russell's Circumplex Model of Affect allow emotions to be represented in terms of valence and arousal, providing a structured approach for mapping sensory inputs to emotional experiences.

Technological advancements such as the Web Serial API have enabled real-time communication between web applications and hardware devices, facilitating the implementation of haptic feedback systems. These developments create opportunities for integrating sensory-based learning experiences into digital platforms.

However, current educational systems for hearing-impaired learners rarely incorporate multi-sensory approaches, limiting their ability to provide holistic and engaging learning environments.

1.1.6 Survey Review

To further understand the challenges faced by hearing-impaired learners and evaluate the effectiveness of existing solutions, a structured survey was conducted targeting students, educators, and individuals familiar with Sinhala Sign Language. The survey aimed to identify limitations in current learning systems and gather user expectations for an improved platform.

A total of **50 respondents** participated in the survey, including hearing-impaired students. The collected responses provide valuable insights into accessibility challenges, learning preferences, and technological requirements.

A. Need for Interactive Gesture-Based Learning

A significant proportion of respondents (82%) expressed a strong preference for interactive learning systems that allow them to practice and evaluate sign language gestures in real time. Only 18% preferred traditional video-based learning without feedback.

This highlights the importance of integrating **gesture recognition and evaluation mechanisms**, enabling users to actively engage in the learning process rather than passively consuming content.

B. Importance of Lip-Reading and Speech Support

Approximately 74% of respondents indicated that lip-reading assistance would significantly improve their communication skills. Many participants emphasized the difficulty of understanding speech without visual cues, particularly in online learning environments.

This demonstrates the necessity of integrating **visual speech recognition tools** to complement sign language learning.

C. Role of Emotion and Engagement in Learning

Survey results revealed that 69% of respondents struggle to maintain attention in online learning environments. Furthermore, 61% stated that systems capable of adapting to their emotional state would enhance their learning experience.

These findings reinforce the importance of incorporating **emotion-aware mechanisms** to improve engagement and learning outcomes.

D. Interest in Multi-Sensory Learning Approaches

More than 65% of respondents expressed interest in alternative learning methods such as vibration-based feedback and visual representations of sound. This indicates that multi-sensory learning can play a significant role in enhancing accessibility and engagement.

Survey Summary

The survey results clearly indicate that existing systems fail to meet the diverse needs of hearing-impaired learners. Users require:

- Interactive and **practice-based learning environments**
- Real-time **gesture evaluation and feedback**
- Support for **lip-reading and speech visualization**
- Emotion-aware systems to improve **engagement**
- Multi-sensory features for **enhanced learning experiences**

These insights strongly justify the development of an integrated platform that combines multiple assistive technologies into a single cohesive system.

1.2 Research Gap

The review of existing literature and available tools reveals a cluster of critical and interrelated gaps in the current landscape of educational technology for hearing-impaired students in Sri Lanka. These gaps collectively motivate the development of the HearMe platform.

First, no existing SSL-specific self-learning platform provides automated real-time gesture correctness evaluation. Existing tools either perform translation without feedback, require instructor involvement, or address sign languages other than SSL. No existing platform integrates a dual-format sign demonstration combining real-world reference video with three-dimensional avatar rendering, despite pedagogical evidence supporting multiple means of representation. Existing LSTM-based SSL recognition systems do not incorporate a CNN sub-classifier to reinforce frame-level accuracy for static signs, and no existing system implements per-gesture feedback for compound multi-gesture SSL signs.

Second, there is an absence of a Sinhala-specific lip-reading assistant designed for hearing-impaired learners rather than for general speech technology research. Most public lip-reading datasets are built around English phonetics and do not represent Sinhala pronunciation patterns, syllable structures, or culturally familiar learning contexts. Existing tools tend to present speech as a one-way demonstration without structured guidance on what aspect of the learner's imitation is correct or incorrect. There is no accessible avatar-based playback system with multiple viewing modes adapted for Sinhala language education.

Third, existing e-learning platforms for hearing-impaired users do not incorporate real-time emotional state monitoring or adaptive intervention mechanisms, resulting in passive learning experiences with limited personalisation. No existing system integrates emotion detection, attention monitoring, and adaptive Sign Language quiz generation within a unified, web-based architecture accessible without specialised hardware. Teacher-facing dashboards in existing educational platforms lack per-student emotional analytics.

Fourth, despite progress in individual components including MER systems, speech-to-text transcription, haptic devices, and accessible design, no integrated platform exists that combines all of these elements in a web-native, e-learning-oriented application specifically designed for hearing-impaired students. The integration of AI-powered conversational guidance for contextual music explanation represents an unexplored direction in accessible music technology, particularly significant in educational contexts where understanding, not just perception, is the learning objective.

Add research gap table

1.3 Research Problem

The central research problem addressed by this dissertation is the comprehensive educational exclusion of hearing-impaired students from key dimensions of the digital learning experience in Sri Lanka. Specifically, hearing-impaired students lack access to an integrated, AI-powered e-learning platform that simultaneously addresses sign language acquisition with automated gesture feedback, visual pronunciation training through culturally appropriate lip-reading assistance, emotionally intelligent adaptive engagement monitoring, and multi-sensory music accessibility. This problem may be formally stated as follows:

How can a comprehensive, web-based, AI-powered inclusive e-learning platform be designed and implemented to effectively address the sign language learning, pronunciation development, emotional engagement, and music accessibility needs of hearing-impaired students in Sri Lanka, thereby reducing educational barriers, enhancing learning engagement, and improving learning outcomes through the integration of deep learning, computer vision, natural language processing, and accessible hardware?

This problem encompasses interconnected sub-problems across four domains: the design of a real-time SSL gesture evaluation system using deep learning; the development of a Sinhala-specific visual speech recognition and pronunciation guidance system; the creation of an emotion-adaptive engagement monitoring and intervention system for hearing-impaired learners; and the development of a multi-sensory music experience that translates audio into visual, tactile, and conversational channels accessible without hearing.

1.4 Research Objectives

The following research objectives guide the design, implementation, and evaluation of the HearMe platform:

Objective 1: To design and implement an Interactive SSL Self-Learning and Gesture Evaluation Module that provides hearing-impaired students with automated real-time gesture correctness feedback, a dual-format sign demonstration system, and per-gesture evaluation for compound signs, using MediaPipe Holistic landmark extraction and a hybrid LSTM and CNN deep learning pipeline.

Objective 2: To design and implement an AI-Powered Sinhala Lip-Reading Assistant with 3D Avatar Feedback that demonstrates Sinhala mouth movements through an avatar and supports visual pronunciation learning through a CNN-LSTM temporal sequence model trained on a custom Sinhala-specific lip-movement dataset.

Objective 3: To design and implement a Teacher-Student Engagement Module with Emotion-Adaptive Learning that detects student emotional states in real time through the DeepFace facial expression recognition library, monitors attention, triggers adaptive Sign Language quiz interventions when disengagement is detected, and provides teachers with actionable emotional analytics through a comprehensive dashboard.

Objective 4: To design and implement a Multi-Sensory Music Experience Platform (Resonance) that translates audio music into simultaneous visual, tactile, and conversational channels through CNN-based music emotion recognition, real-time speech-to-text transcription, a custom ESP32-based haptic feedback device, and an AI-powered contextual music guide.

Objective 5: To integrate the four modules into a unified, cohesive platform architecture with a shared user identity layer, consistent terminology, and a common design language that ensures accessibility for hearing-impaired users without specialist hardware requirements.

Objective 6: To evaluate the platform through functional testing, performance benchmarking, user acceptance testing with hearing-impaired participants, and system integration testing, validating accuracy, usability, and educational effectiveness across all four modules.

METHODOLOGY

The methodology of the HearMe platform is organised as an integrated research-and-prototype development workflow spanning four interconnected modules. Each module addresses a distinct dimension of educational exclusion for hearing-impaired students and employs a tailored technical approach appropriate to its specific requirements. The development process follows an iterative, agile-inspired methodology that balances empirical machine learning experimentation with software engineering practice and user-centred design validation.

2.1 Overall System Architecture

The HearMe platform is implemented as a distributed, full-stack web application comprising four semi-independent module systems that share a common user identity layer and a consistent design language. The overall architecture follows a microservice-inspired pattern in which each module maintains its own frontend components and backend services while sharing common infrastructure including authentication, user profile management, and cross-module navigation.

The platform's architecture can be characterised as a four-tier system. The presentation tier consists of React-based frontend applications for each module, unified under a common navigation shell. The application tier consists of module-specific backend services implemented in Flask (sign language and lip-reading modules) and Node.js with Express (emotion and engagement module), with FastAPI microservices for computationally intensive inference tasks (music emotion recognition and transcription). The inference tier consists of trained TensorFlow/ Keras deep learning models served through the application backends. The data tier consists of MongoDB for session and analytics data, static file storage for reference videos and avatar assets, and in-memory session state for real-time inference pipelines.

The four modules communicate with each other through a shared user session context that stores the learner's identity, progress data, and cross-module preferences. This context enables the platform to present a coherent learning experience, such as remembering which SSL vocabulary the learner has studied, which lip-reading stages have been completed, what emotional patterns have been observed, and what music sessions have been experienced.

System Architecture Diagram Description: The overall HearMe system architecture comprises a shared React frontend shell that routes users between the SSL Learning Module (React frontend with Vite, communicating via HTTP to a Flask REST API backed by TensorFlow/Keras LSTM and CNN models), the Lip-Reading Assistant (React frontend communicating with a Flask backend serving the CNN-LSTM lip recognition model and the GLB-based 3D avatar renderer), the Emotion-Adaptive Engagement Module (React frontend backed by a MERN stack with a Python Flask microservice serving the DeepFace emotion detection and NLP quiz generation pipelines), and the Resonance Music Platform (React frontend with Web Audio API and Web Serial API for haptic control, communicating with a FastAPI inference backend for CNN-based emotion prediction and a separate FastAPI transcription backend connecting to Google Cloud Speech-to-Text). All modules share a MongoDB Atlas database for user profiles, progress records, and analytics data, and communicate with the shared authentication layer via REST API endpoints.

Subsystem	**Technology Stack**	**Responsibility**
SSL Gesture Module Frontend	React 18, Vite 5, MediaPipe Holistic JS	Webcam capture, landmark extraction, dual-format sign demo, feedback display
SSL Gesture Module Backend	Python, Flask, TensorFlow/ Keras	LSTM and CNN model inference, evaluation logic, gesture scoring API
Lip-Reading Assistant Frontend	React 18, GLB 3D avatar renderer	Avatar playback, webcam lip capture, gamified progress dashboard

Subsystem	Technology Stack	Responsibility
Lip-Reading Assistant Backend	Python, Flask, TensorFlow/ Keras	CNN-LSTM lip recognition inference, phoneme-to-viseme mapping, feedback API
Emotion-Adaptive Module Frontend	React 18, WebRTC, DeepFace JS	Real-time webcam emotion display, teacher dashboard, adaptive quiz UI
Emotion-Adaptive Module Backend	Node.js, Express, Python Flask, MongoDB	DeepFace emotion classification, NLP quiz generation, analytics aggregation
Resonance Music Frontend	React 18, Web Audio API, Web Serial API	Audio visualisation, haptic control, chatbot UI, playback summary
Resonance Inference Backend	Python, FastAPI, TensorFlow, librosa	CNN mel-spectrogram inference for valence/arousal prediction
Resonance Transcription Backend	Python, FastAPI, Google Cloud Speech	Speech-to-text, language detection, Llama 3.1 summarisation

Table 2.1: System Subsystem Overview — HearMe Platform Four Modules with Technologies and Responsibilities

2.2 Sign Language Learning Module — Methodology

2.2.1 Research Design and Data Collection

The SSL module follows an iterative prototyping methodology consistent with agile software development principles. Development proceeded through four phases: requirements analysis with input from hearing-impaired community members; prototype development with landmark extraction and initial model training; iterative refinement based on evaluation results and user feedback; and final integration and testing. The SSL400 dataset (Abhishek, 2023), comprising 400 Sri Lankan Sign Language word classes represented as video sequences recorded by multiple signers under varied conditions, serves as the primary vocabulary reference.

A vocabulary categorisation framework is established to guide model selection and evaluation logic. Category 1 encompasses static and simple dynamic signs such as colour terms, numerals, and common greetings, which are evaluated using the CNN sub-classifier pipeline. Category 2 encompasses complex dynamic signs involving sequential multi-gesture production, such as kinship terms and common action words, which are evaluated using the LSTM temporal sequence classifier. For the initial implementation, ten to fifteen vocabulary classes are selected from each category to maximise inter-class distinctiveness and model accuracy.

2.2.2 Landmark Extraction and Sequence Normalisation

For each video clip in the SSL400 dataset, MediaPipe Holistic is applied frame by frame to extract three groups of landmarks: 33 pose landmarks representing the full body skeleton, 21 left hand landmarks, and 21 right hand landmarks. Each landmark includes x, y, and z coordinates normalised relative to the image dimensions. The three landmark groups are concatenated per frame to produce a composite feature vector. The pose landmark group contributes 132 features (33 landmarks times 4 values including visibility). Each hand landmark group contributes 63 features (21 landmarks times 3 coordinate values). The full MediaPipe Holistic output yields 1,662 features per frame when all landmark groups are included at the extraction layer.

All landmark sequences are normalised to a fixed length of 30 frames. For sequences longer than 30 frames, uniform temporal sub-sampling selects 30 evenly spaced frames. For sequences shorter than 30 frames, zero-padding extends the sequence to 30 frames. The choice of 30 frames was determined empirically; at a typical webcam capture rate of 15 to 30 frames per second, 30 frames correspond to a temporal window of approximately one to two seconds, sufficient to capture the temporal extent of most SSL signs in the SSL400 dataset. A custom CNN dataset is also collected for static sign recognition using OpenCV, capturing approximately 300 to 400 greyscale images per class at 64 by 64 pixel resolution under controlled conditions.

Data Preprocessing Pipeline Description: The preprocessing pipeline begins with raw video frame capture from either the SSL400 dataset or the live webcam feed. MediaPipe Holistic processes each frame to extract pose, left hand, and right hand landmarks. These are concatenated into a 1,662-dimensional feature vector per frame. For the LSTM model, 30 consecutive frame vectors are assembled into a sequence matrix of shape (30, 1662), with temporal sub-sampling applied for longer sequences and zero-padding applied for shorter sequences. For the CNN model, individual frames are converted to greyscale, cropped to the hand region, and resized to 64 by 64 pixels before being passed through the convolutional classification pipeline.

2.2.3 Model Architecture and Training

The primary sign recognition model is a sequential LSTM network. The architecture comprises three LSTM layers followed by two Dense layers. The first LSTM layer contains 128 units with return sequences set to true. The second LSTM layer contains 256 units, also with return sequences set to true. The third LSTM layer contains 128 units with return sequences set to false, collapsing the sequence into a fixed-length representation. This representation is passed to a Dense layer of 128 units with ReLU activation, followed by a final Softmax Dense layer whose output dimension equals the number of vocabulary classes. The model is compiled with the Adam optimiser and categorical cross-entropy loss. Training runs for up to 2000 epochs with early stopping applied when validation accuracy ceases to improve for 50 consecutive epochs, and a model checkpoint callback saves the weights corresponding to peak generalisation performance.

The CNN sub-classifier architecture for static sign recognition comprises two convolutional-pooling blocks followed by a fully connected classification head. The first convolutional layer applies 32 filters of size 3 by 3 pixels with ReLU activation. A MaxPooling layer with a 2 by 2 pool size follows. The second convolutional layer similarly applies 32 filters with ReLU activation, followed by a second MaxPooling layer. The flattened output is passed through a Dense layer of 128 units with ReLU activation and a final Softmax Dense layer. The model is compiled with the Adam optimiser and categorical cross-entropy loss, training with an 80:20 training-validation split for up to 200 epochs.

2.2.4 Real-Time Detection Pipeline and Evaluation Logic

The real-time detection pipeline processes the live webcam feed captured in the browser. The webcam feed is accessed via the browser MediaDevices API, and each captured frame is passed to the MediaPipe Holistic library running in the browser to extract the full set of landmarks. A sliding sequence buffer of 30 frames is maintained in JavaScript state. When the buffer reaches the target length, the sequence is serialised and dispatched to the Flask backend via an asynchronous HTTP POST request to the /predict endpoint. The Flask backend deserialises the sequence, passes it to the LSTM model for inference, and returns the predicted class and confidence score. A majority voting smoothing mechanism with a history buffer of 10 predictions reduces prediction noise from transient frames and momentary occlusions.

The gesture evaluation logic implements differentiated feedback mechanisms for Category 1 and Category 2 signs. For Category 1 signs, the evaluation logic tracks the number of attempts per sign and awards decreasing score credit per additional attempt: full credit on the first correct attempt, partial credit on the second, and minimal credit on the third. Exceeding three attempts marks the sign for further practice. For Category 2 compound signs, each constituent gesture within the multi-gesture sequence is evaluated independently, providing per-gesture feedback that identifies which specific sub-gestures within the compound sign require correction.

2.3 Lip-Reading Assistant — Methodology

2.3.1 Research Design and Dataset Creation

The lip-reading assistant methodology is organised as a research-and-prototype workflow beginning with problem scoping, continuing through dataset design and model development, and ending with prototype evaluation and feedback synthesis.

The research design is applied and iterative, combining computer vision, sequence modelling, animation, and human-centred instructional design. Because this module is a Data Science specialisation project, the methodology gives equal weight to the learning problem and the data process.

The dataset is the foundation of the component. A controlled data collection protocol is proposed in which participants articulate selected Sinhala words and short sentences in front of a camera under stable lighting conditions. The recording setup emphasises a frontal view, consistent distance, and unobstructed visibility of the mouth area. The custom dataset is designed with two related units of annotation: the word or sentence label, allowing the model to associate a full utterance with a reference sequence, and the phoneme or visual speech label, supporting more granular feedback. Because existing lip-reading datasets do not adequately represent Sinhala phonemes and sentence structures, a proprietary Sinhala lip-movement dataset is created for the module.

Table 2.3: Data Collection and Preprocessing Specifications — Sinhala Lip-Reading Dataset

Stage	Specification	Purpose	Output
Recording	Frontal camera, consistent distance, controlled lighting	Reduce variation in face pose and exposure	Raw video clips
Segmentation	Split utterances into frame sequences	Prepare temporal learning samples	Sequence folders
Landmark extraction	Detect lip and facial landmarks using MediaPipe	Identify the lip ROI accurately	Normalised coordinates
Normalisation	Resize and align lip crop to fixed dimensions (128x128)	Standardise model input	Normalised frame sequences
Labeling	Word-level and phoneme-level annotations	Enable fine-grained feedback	Training labels
Split design	Train/validation/test separation by speaker	Prevent overfitting and leakage	Model-ready dataset

2.3.2 Preprocessing and CNN-LSTM Model Design

Once raw clips are captured, preprocessing converts them into a standardised format for learning. A landmark detector identifies the face outline and the points around the lips so that the mouth region can be isolated consistently across frames. The ROI is cropped, resized, and normalised so that every frame has the same spatial dimensions and a similar numerical range. Each utterance is represented as a sequence of frames in correct temporal order. A fixed-length or padded sequence strategy ensures that each example is comparable without discarding information. Minor head motion, lighting changes, or incomplete face visibility are handled through image normalisation and sequence filtering.

The model combines convolutional feature extraction with long short-term memory sequence learning. The CNN extracts local spatial features from each lip frame — mouth openness, lip curvature, upper and lower lip boundary relations — and passes these to the LSTM, which captures temporal progression across the utterance. The CNN-LSTM architecture allows the system to distinguish between isolated but visually similar mouth positions and full sequences with different articulatory meaning. The output layer maps the sequence into the closest target category or phoneme group, with dropout regularisation and early stopping applied to reduce overfitting.

2.3.3 Phoneme-to-Viseme Mapping and Avatar Feedback

Phoneme-to-viseme mapping is the bridge between linguistic theory and visual feedback. A phoneme is an abstract sound unit while a viseme is its observable mouth shape. In Sinhala, some sounds map to similar lip shapes while others have more distinctive visible cues. The mapping table groups sounds into visually meaningful categories such as open mouth, rounded lips, closed mouth, and stretched mouth.

Viseme Group	**Sinhala Examples**	**Visual Cue**	**Feedback Use**
Open mouth	அ, ஆ, இ	Wide vertical opening	Prompt learner to open jaw wider
Rounded lips	ஓ, ஓ, உ	Circular or forward lip shape	Prompt learner to round lips
Closed lips	ம, ப, ப	Full lip closure	Prompt learner to close lips fully
Stretched lips	இ, ஈ	Horizontally stretched shape	Prompt learner to relax rounding
Neutral transition	Short connectors	Minimal visible change	Suggest smooth timing and pacing

Table 2.4: Phoneme-to-Viseme Mapping used by the Avatar Feedback Engine

The avatar rendering layer presents ideal lip motion for each target phoneme or word. The system supports adjustable playback speed and multiple viewing angles so that learners can inspect how the lips round or how the jaw changes from one phoneme to another. Each target pronunciation sequence is aligned with visual events in the avatar through timing markers or a sequence of viseme frames that the renderer plays in order. The progress scoring layer converts repeated attempts into a learning record, awarding credit for improvement relative to prior attempts so that the score functions as a motivational signal rather than a final judgment.

2.4 Emotion-Adaptive Engagement Module — Methodology

2.4.1 System Architecture and Emotion Detection

The Teacher-Student Engagement Module with Emotion-Adaptive Learning is developed following the Agile software development framework, organised into four primary sprints addressing: Emotion Detection and Attention Monitoring; Adaptive Learning Intervention Engine; NLP-based Quiz Generation Pipeline; and Teacher Dashboard and Analytics. The system architecture follows a three-tier MERN stack design comprising a React.js frontend, a Node.js and Express.js backend, a Python Flask microservice for machine learning inference, and a MongoDB database layer.

The emotion detection subsystem uses the DeepFace library, which utilises ensemble methods across multiple pre-trained models including VGG-Face, FaceNet, OpenFace, and ArcFace. The system continuously monitors student facial expressions through webcam inputs during video lectures and live sessions, classifying emotional states into five categories: Happy, Neutral, Confused, Bored, and Disengaged. An emotion state machine governs transitions between emotional states, requiring sustained detection over multiple frames before triggering a state change to prevent spurious responses to momentary expressions. The system achieves approximately 97% face detection accuracy and 80% emotion classification accuracy in controlled settings.

The attention detection mechanism uses HOG features and facial landmark tracking to identify head pose deviations indicating inattentiveness. When inattention is detected for a sustained duration, the system automatically pauses lesson video content and displays an attention reminder animation, resuming automatically upon focus restoration.

2.4.2 Adaptive Intervention Engine and NLP Quiz Generation

The adaptive learning intervention engine monitors the duration of emotional states and triggers contextually appropriate interventions when prolonged disengagement or confusion is detected. Two intervention types are implemented. The Sign Language Quiz Intervention presents an SSL vocabulary quiz drawn from the current lesson content, reinforcing learning through a visual and interactive activity that re-engages the student. The Visual Matching Exercise Intervention presents image-to-sign matching tasks that require active cognitive engagement without requiring audio processing. Both interventions are designed to be contextually relevant to the lesson being studied and to resume the lesson flow smoothly upon completion.

The NLP-based quiz generation pipeline operates in three stages. The transcript generation stage produces a text transcript from video lecture audio using automatic speech recognition. The keyword extraction stage applies NLP techniques including

TF-IDF weighting and named entity recognition to identify the most educationally salient terms in the transcript. The question generation stage constructs quiz questions from the extracted keywords and their surrounding context using template-based and transformer-informed generation strategies. The generated questions are formatted for visual presentation, emphasising image-based and matching formats appropriate for hearing-impaired learners.

2.4.3 Teacher Dashboard

The Teacher Dashboard provides instructors with real-time and aggregated emotional analytics, individual student attention reports, and engagement metrics to enable informed pedagogical decisions. The dashboard displays a live emotional state timeline for each student currently attending an online session, a per-student attention report summarising the total duration of attentive versus inattentive periods, and class-wide engagement metrics aggregated across all students in a session. Historical session reports enable instructors to review engagement trends over time and identify vocabulary categories or lesson segments that consistently produce confusion or disengagement. All analytics are anonymised in compliance with data protection principles and are accessible only to the supervising instructor.

2.5 Multi-Sensory Music Experience — Methodology

2.5.1 System Architecture and Data Acquisition

The Resonance platform is built on a decoupled microservice architecture with four primary subsystems: the Emotion Recognition Server, the Transcription Server, the React Frontend Application, and the ESP32 Hardware Module. These subsystems communicate via well-defined REST API contracts, enabling independent development and testing of each component. The user loads an audio file into the React frontend, which decodes it into an AudioBuffer. The frontend simultaneously extracts audio chunks for emotion analysis and transcription, renders real-time visualisations using the Web Audio API, and dispatches haptic commands to the ESP32 via the Web Serial API.

The emotion recognition model is trained on the DEAM (Database for Emotional Analysis in Music) dataset, containing 1,802 songs annotated with continuous Valence and Arousal values on a 1 to 9 scale. Audio files in various formats are loaded using librosa and resampled to a standardised 44,100 Hz mono format. Each audio track is segmented into 5-second chunks of 220,500 samples, with partial segments longer than 2.5 seconds zero-padded to the required length. The dataset is split by Song ID to prevent data leakage: Training Set at 70%, Validation Set at 15%, and Test Set at 15%.

Table 2.5: Mel-Spectrogram Computation Parameters

Parameter	Value	Description
n_mels	128	Number of mel filter bank bands
n_fft	2048	FFT window size in samples
hop_length	512	Stride between FFT frames in samples
Sample Rate	44,100 Hz	Input signal sample rate
Segment Length	5.0 seconds	Duration of each analysis window
Input Shape	(128, 431, 1)	CNN input after spectrogram extraction and reshape

2.5.2 CNN Model Architecture for Music Emotion Recognition

The CNN is designed for regression tasks, predicting continuous valence and arousal scores. The architecture consists of four convolutional blocks followed by fully connected layers. Each convolutional block contains a Conv2D layer for spatial feature extraction from the spectrogram, MaxPooling2D for spatial dimension reduction, BatchNormalisation for learning stabilisation, and Dropout for overfitting prevention. A Flatten layer converts the 2D feature maps to 1D vectors for subsequent dense layers, which learn high-level representations through ReLU activation. The output layer consists of 2 neurons with linear activation, predicting Valence and Arousal scores. The model is trained using the Adam optimiser and MSE loss function, with early stopping and data augmentation applied. Mel-spectrograms are computed with 128 mel filter bands, an FFT window of 2048 samples, and a hop length of 512 samples.

2.5.3 Haptic Feedback System and Hardware Integration

The haptic feedback subsystem translates audio frequency data into physical vibrations through a custom ESP32-based device. The frontend implements a beat-detection algorithm using a frame-to-frame energy delta approach. For each audio frame, RMS energy is computed across a configurable frequency bin range. When the energy delta between consecutive frames exceeds a threshold of 0.005, a beat onset is detected and the motor output is set to currentEnergy times 255 times 2.5, capped at 255. When no onset is detected, output decays exponentially by 25% per frame, creating a natural vibration fade-out. The ESP32 receives motor commands via a text-based serial protocol, with left and right motor intensities transmitted as a CSV pair terminated by a newline. Two PWM channels are configured on GPIO pins 18 and 5 at 1 kHz with 8-bit resolution.

The Web Serial API connection enables browser-native hardware communication without native drivers, companion applications, or browser extensions, managed at approximately 30 fps to prevent serial buffer overflow. This represents a novel application of the Web Serial API in accessible music systems, eliminating the adoption barriers associated with Bluetooth pairing and native driver installation.

2.5.4 Transcription System and AI Music Guide

The Transcription Backend handles audio format conversion and Google Cloud Speech-to-Text API communication. Files received via multipart/form-data are processed using FFmpeg with strict parameters: audio resampled to 16 kHz optimal for speech recognition, downmixed to mono, and encoded as 16-bit signed little-endian PCM. The transcription agent polls the transcription endpoint every 1000 ms with 10-second audio chunks using a sliding window approach. For Sinhala language processing, a character-level Unicode validation step ensures that at least 70% of non-whitespace characters fall within the Sinhala Unicode range U+0D80 to U+0DFF, preventing false-positive romanised outputs.

The AI Music Guide chatbot employs a contextual injection pattern: real-time transcription history, current emotion data (valence, arousal, emotion category), playback position, and playback summary are dynamically embedded in the Groq Llama 3.1 system prompt for each user query. This creates a genuinely context-aware educational companion that can explain the emotional, structural, and lyrical significance of music without requiring domain-specific fine-tuning. The Playback Summary Report engine aggregates session emotion and transcription data into structured learning summaries, computing dominant emotion, peak energy timestamp, average valence and arousal values, and a Llama 3.1-generated narrative summary of the musical journey.

2.6 Platform Integration and Data Flow

The four HearMe modules are integrated through a shared user session architecture. A common authentication layer manages user registration, login, and session token distribution across all module frontends. The shared session context stores the learner's identity, module-specific progress records, and cross-module preferences. This enables the platform to present a coherent learning journey, such as the emotion-adaptive module acknowledging that a learner has completed certain SSL vocabulary units and adjusting quiz intervention content accordingly.

Data flow within each module follows a consistent pattern: the frontend captures user input (webcam video, audio file, or user interaction), extracts features locally where computationally feasible (MediaPipe landmark extraction in the browser for the SSL and lip-reading modules, Web Audio API processing for the music module), serialises the extracted features or audio chunks, and dispatches them to the module-specific backend via asynchronous HTTP REST API calls. The backend performs inference, applies evaluation or feedback logic, and returns structured JSON responses that the frontend uses to update the display state.

Module Interaction Diagram Description: The four HearMe modules interact through the shared user session layer. The SSL Learning Module and the Lip-Reading Assistant share the MediaPipe Holistic framework for landmark extraction, enabling shared utility functions and consistent preprocessing logic. The Emotion-Adaptive Module monitors the learner's emotional state throughout all learning activities and may surface cross-module insights to the Teacher Dashboard. The Resonance Music Platform operates as a standalone experience module but contributes session analytics to the shared user profile. All modules write progress and analytics records to the shared MongoDB database, enabling cross-module progress reporting.

2.7 Commercialisation Aspects

The commercialisation of the HearMe platform is best understood as educational deployment for the inclusive technology market rather than as consumer entertainment. The primary target customers are special education schools, universities with disability support programmes, rehabilitation centres, NGOs working in disability support, and government ministries of education in South Asia. A freemium model provides access to a limited vocabulary set, basic emotion monitoring, and fundamental music accessibility features at no cost, with a premium subscription tier unlocking the full SSL400 vocabulary, advanced analytics, personalised progress tracking, and the haptic hardware integration.

An institutional licensing pathway targets bulk deployment across large student populations without requiring individual subscription management. Revenue stability under this model is supported by annual licence fees calibrated to institution size. The platform's modular architecture enables content customisation, where institutions can adapt vocabulary sets, avatar recordings, quiz content, and feedback scripts for specific learner populations without requiring complete redevelopment. The SSL landmark extraction pipeline, LSTM model, and evaluation logic are language-agnostic, enabling adaptation to Indian Sign Language, Pakistani Sign Language, and other regional sign languages through dataset substitution and model retraining, significantly expanding the addressable market.

Mobile deployment via Android and iOS applications represents an important commercialisation direction, particularly for South Asian markets where smartphone-first digital engagement is prevalent. The Flask and FastAPI REST API backends are mobile-client-compatible, and the React frontend code can be adapted for React Native with moderate development effort. Cloud deployment on AWS or Google Cloud enables horizontal scaling to support institutional customer loads. All commercial deployments must respect ethical constraints related to user data privacy: the system handles face video, lip sequences, webcam streams, and learner performance data, all of which are sensitive. Commercial deployment requires consent forms, secure storage, clear privacy policies, and compliance with Sri Lanka's Online Safety Act and relevant data protection guidelines.

2.8 Testing and Implementation

2.8.1 Testing Strategy

The platform testing strategy encompasses unit testing, integration testing, system testing, and user acceptance testing across all four modules. Unit testing validates individual functional components in isolation: the landmark extraction and sequence normalisation pipelines, the model inference endpoints, the evaluation and scoring logic, the emotion detection and attention monitoring algorithms, and the haptic motor command generation. Integration testing verifies the correct interaction between the frontend, backend, and model inference layers within each module, validating complete user flows from input capture through feedback display.

System testing validates inter-module data sharing, cross-module navigation, and the integrity of the shared user session architecture. User acceptance testing is conducted with hearing-impaired participants recruited through collaboration with disability support organisations in Sri Lanka, evaluating classification accuracy, real-time performance, usability, and user-reported utility and accessibility.

2.8.2 Implementation Overview

The SSL module frontend is implemented in React 18 with Vite bundling, using the MediaPipe Holistic JavaScript library for browser-based landmark extraction. The Flask backend loads the trained TensorFlow/Keras LSTM and CNN models at application startup and holds them in memory to minimise inference latency. The lip-reading assistant is implemented with a similar React and Flask architecture, with the GLB-based 3D avatar model hosted as a browser-rendered WebGL component. The emotion-adaptive module is implemented using the MERN stack, with a Python Flask microservice handling DeepFace inference and NLP quiz generation. The Resonance platform is implemented with React 18 and Vite for the frontend, two FastAPI backends for emotion inference and transcription respectively, and C++ Arduino firmware for the ESP32 haptic device programmed using PlatformIO.

Layer	Technologies
Frontend	React 18, Vite 5, Axios, Lucide-React, Web Audio API, Web Serial API, MediaPipe Holistic JS, Three.js (avatar)
Backend	Python 3.10, Flask, FastAPI, Uvicorn ASGI, Node.js, Express.js
Machine Learning	TensorFlow/Keras, librosa, NumPy, Pandas, Scikit-learn, DeepFace
External APIs	Google Cloud Speech-to-Text, Groq Cloud Llama 3.1
Database	MongoDB Atlas
Hardware	ESP32, PlatformIO, Arduino C++, ERM haptic motors

Table 2.6: Technologies Used across the HearMe Platform

CHAPTER 3: RESULTS AND DISCUSSION

This chapter presents the quantitative performance results and qualitative evaluation findings from the HearMe platform across all four modules. Results are organised by module, covering classification performance, system response metrics, user acceptance testing outcomes, and cross-module findings. Research findings are then synthesised and discussed in terms of the research objectives and the broader contribution to inclusive educational technology. The chapter concludes with a summary of each student's individual contribution.

3.1 Results

3.1.1 Sign Language Module — Classification Results

The CNN pipeline for Category 1 static sign recognition achieved strong classification performance across all tested vocabulary classes. The highest classification accuracy of 100% was achieved for the numerals 1 through 10 vocabulary class, reflecting the high inter-class distinctiveness of numeral signs in SSL, where each numeral involves a qualitatively distinct hand configuration. Colour terms achieved 94.44% accuracy. Common greetings achieved 96.00% accuracy, demonstrating robust performance for a vocabulary class of high practical utility.

Vocabulary Class	No. of Classes	Model	Accuracy (%)	F1 Score
Colour terms	9	CNN	94.44	0.94
Numerals 1–10	10	CNN	100.00	1.00
Common greetings	5	CNN	96.00	0.96
Basic shapes	6	CNN	91.67	0.92
Simple action signs	8	CNN	93.75	0.93

Table 3.1: Category 1 SSL Classification Accuracy and F1 Scores by Vocabulary Class

The LSTM model for Category 2 complex dynamic sign recognition achieved high accuracy across tested vocabulary classes. The family members class, comprising three sign classes, achieved 100% classification accuracy. The light colour family class of eleven classes achieved 95.45% accuracy. Common action verbs achieved 97.14% accuracy across seven classes.

Vocabulary Class	No. of Classes	Sequence Length	Accuracy (%)	F1 Score
Family members	3	30 frames	100.00	1.00
Light colour family	11	30 frames	95.45	0.96
Common action verbs	7	30 frames	97.14	0.97
Emotion terms	6	30 frames	91.67	0.92
Daily activity signs	9	30 frames	88.89	0.89

Table 3.2: Category 2 SSL Classification Accuracy and F1 Scores by Vocabulary Class

The majority voting smoothing mechanism was evaluated for its impact on prediction consistency in real-time operation. A smoothing window of 10 frames achieves 94.1% prediction consistency at 420 milliseconds mean latency, representing the optimal trade-off between accuracy and interactive responsiveness.

Condition	Consistency Rate (%)	Mean Latency (ms)
Without smoothing	74.3	210
Smoothing (window = 5)	87.6	315
Smoothing (window = 10)	94.1	420
Smoothing (window = 15)	95.8	630

Table 3.3: Effect of Majority Voting Smoothing Window Size on Prediction Consistency and Latency

3.1.2 Sign Language Module — User Acceptance Testing

User acceptance testing was conducted with twelve hearing-impaired participants recruited through disability support organisations in Sri Lanka. Participants rated the clarity of the reference video demonstration most highly at 4.58 out of 5. The usefulness of per-gesture feedback for Category 2 signs received the highest individual rating of 4.67. Overall satisfaction with the system received a mean score of 4.50.

Evaluation Criterion	Mean Score (1-5)	Std Deviation
Ease of use of the interface	4.42	0.51
Clarity of reference video demonstration	4.58	0.49
Utility of the 3D avatar demonstration	4.33	0.65
Clarity of gesture correctness feedback	4.25	0.72
Usefulness of per-gesture feedback (Category 2)	4.67	0.49
Overall satisfaction with the system	4.50	0.52

Table 3.4: User Acceptance Testing Results for the SSL Module — Mean Scores on 1-5 Likert Scale (n=12)

3.1.3 Emotion-Adaptive Module — Performance Results

The DeepFace-based emotion detection system achieved approximately 97% face detection accuracy and 80% overall emotion classification accuracy. Classification accuracy varied by emotional state, with Happy and Neutral states achieving the highest accuracy due to their more distinct and socially common facial configurations. Confused and Disengaged states showed slightly lower accuracy, reflecting the subtler and more varied facial cues associated with these states.

Emotional State	Detection Accuracy (%)	Notes
Happy	89.2	Highest accuracy; distinctive smile and eye crinkling features

Emotional State	Detection Accuracy (%)	Notes
Neutral	85.4	High accuracy; serves as baseline reference state
Confused	76.3	Moderate accuracy; furrowed brow and head tilt cues
Bored	72.8	Lower accuracy; subtle postural and gaze cues
Disengaged	74.1	Lower accuracy; overlap with bored state facial features

Table 3.5: Emotion Detection Accuracy by Emotional State Category

The adaptive intervention system demonstrated a significant positive effect on learner re-engagement. When Sign Language quizzes were triggered upon detection of sustained disengagement, 78% of students returned to attentive engagement within the subsequent three-minute window, compared to 31% in a control group without intervention. The NLP quiz generation pipeline produced contextually relevant questions with an 82% acceptance rate from expert evaluators.

3.1.4 Resonance Music Platform — Performance Results

The CNN emotion recognition model was evaluated across four aggregation configurations. The Dynamic Graph Rewiring Attention Aggregator outperformed all other methods across all metrics, showing the fastest decrease in training loss and the evaluation loss remaining close to the training loss, indicating excellent generalisation. This aggregator achieved the highest AUC values across training, validation, and test splits, with the attention mechanism's ability to selectively weight the most relevant mel-spectrogram structural features explaining its advantage over uniform aggregation strategies.

Aggregator	Training AUC	Validation AUC	Test AUC	Generalisation
Sum Aggregator	High	Moderate	Moderate	Minor overfitting observed
Concat Aggregator	Moderate	Variable	Moderate	Instability in generalisation

Aggregator	Training AUC	Validation AUC	Test AUC	Generalisation
Neighbor Aggregator	Lower	Lower	Lower	Poor generalisation
Dynamic Graph Rewiring Attention	Highest	Highest	Highest	Excellent generalisation

Table 3.6: Music Emotion Recognition Aggregator Performance Comparison Summary

System integration testing validated the end-to-end functionality of the Resonance platform. All six primary test cases passed, including CNN emotion recognition accuracy across all four emotion quadrants, haptic motor response synchronisation with beat onsets, real-time transcription in English, Spanish, and Sinhala, Web Serial API stability over a five-minute continuous haptic session, AI chatbot contextual relevance during high-arousal segments, and playback summary report generation accuracy.

3.2 Research Findings

The empirical results across all four modules confirm the following key research findings.

Finding 1: Deep learning-based real-time gesture recognition can achieve classification accuracy between 88.89% and 100% for SSL vocabulary using only a standard consumer webcam, without specialist hardware, gloves, or GPU acceleration requirements. The hybrid LSTM and CNN pipeline with MediaPipe Holistic landmark extraction is effective for both static and dynamic SSL sign classification.

Finding 2: Majority voting smoothing with a 10-frame history buffer reduces real-time prediction noise from 74.3% to 94.1% consistency while maintaining interactive latency below 500 milliseconds, confirming the feasibility of the system for practical self-directed learning use.

Finding 3: Per-gesture feedback for compound SSL signs is the most highly valued feature by hearing-impaired users (mean score 4.67 out of 5), confirming that fine-grained sub-gesture identification addresses a genuine and previously unmet pedagogical need.

Finding 4: DeepFace-based emotion detection achieves practical accuracy for real-time classroom monitoring (approximately 80% overall), with adaptive Sign Language quiz interventions producing a 78% re-engagement rate compared to 31% in controls, demonstrating a significant positive effect on learner engagement.

Finding 5: The Dynamic Graph Rewiring Attention aggregator outperforms sum, concatenation, and neighbour aggregation strategies for music emotion recognition across all evaluation metrics, confirming that selective attention to structurally relevant mel-spectrogram features is beneficial for continuous valence and arousal prediction.

Finding 6: Browser-native haptic hardware integration via the Web Serial API is technically viable for real-time music accessibility, achieving stable serial communication at 30 fps over five-minute continuous sessions without buffer overflow, eliminating the adoption barriers associated with native driver installation and Bluetooth pairing.

3.3 Discussion

3.3.1 Strengths of the Platform

The primary strength of the HearMe platform is its comprehensive and integrated approach to the educational exclusion of hearing-impaired students. Rather than addressing a single dimension of accessibility, the platform simultaneously tackles sign language learning, pronunciation guidance, emotional engagement, and music accessibility within a unified architecture. This integration reflects the complexity of educational exclusion, which is rarely reducible to a single technology fix.

The landmark-based feature extraction approach, using MediaPipe Holistic, is computationally efficient and robust to variations in background, lighting, and signer clothing. The dual-format sign demonstration provides a richer learning experience than single-format systems, addressing the pedagogical principle of multiple means of representation. The per-gesture feedback mechanism for compound SSL signs is a novel contribution that addresses a specific unmet need confirmed by user testing. The phoneme-to-viseme mapping and avatar-based pronunciation guidance translate an abstract linguistic concept into a visual learning aid accessible to hearing-impaired students who lack auditory feedback channels.

The Resonance platform's approach to music accessibility — combining real-time emotion visualisation, multi-language transcription, browser-native haptic feedback, and AI-powered contextual guidance — represents a holistic solution that goes beyond the fragmented and partial tools previously available. The AI Music Guide's contextual injection pattern, embedding real-time session data into the LLM system prompt, enables context-aware educational explanations without domain-specific fine-tuning, demonstrating a practical and scalable approach to AI-assisted accessible learning.

3.3.2 Limitations

The primary limitation of the SSL module is the restricted vocabulary size at this stage, with ten to fifteen classes included in the trained model. While sufficient to demonstrate the feasibility and educational value of the approach, this limits practical utility for learners requiring broader SSL vocabulary. Expansion to the full SSL400 vocabulary of 400 classes is the most impactful direction for future development.

The emotion detection system's accuracy is contingent upon adequate webcam quality and lighting conditions, with performance degrading under low illumination or strong backlighting. The system is designed for SSL, limiting direct applicability to other sign language systems without retraining. The lip-reading model's effectiveness depends on the quality and diversity of the recorded Sinhala lip-movement dataset, and the initial prototype necessarily operates with a constrained vocabulary. The haptic feedback device's effectiveness varies across individuals based on vibrotactile perception capability, and users with reduced tactile sensitivity may experience diminished benefit from the haptic channel.

3.3.3 Future Work

Future development priorities include vocabulary expansion for the SSL module to cover the full SSL400 dataset, potentially employing transformer-based architectures that have demonstrated state-of-the-art performance on sequence classification tasks. Mobile application deployment via React Native would substantially expand accessibility, particularly in South Asian markets where smartphone-first digital engagement is prevalent. Model personalisation through transfer learning from individual user data would address signer variability limitations. Gamification enhancements including spaced repetition, achievement systems, and adaptive difficulty selection would improve motivational engagement.

For the lip-reading assistant, future work should focus on expanding the Sinhala lip-movement dataset, extending the phoneme-to-viseme mapping, and integrating the avatar with a more detailed face model for improved articulatory fidelity. For the emotion-adaptive module, expansion of the adaptive intervention content library and exploration of multi-modal emotion detection combining facial expression with text sentiment from sign language inputs would improve intervention quality. For the Resonance platform, integration of haptic rendering for mid-range frequency music content and the development of a mobile haptic companion application represent important accessibility enhancements.

3.4 Summary of Each Student's Contribution

This section explicitly delineates the individual contribution of each team member to the HearMe platform.

Jayasinha J.A.N.Y. (IT22546852) designed and implemented the Interactive SSL Self-Learning and Gesture Evaluation Module in its entirety. This encompasses the vocabulary categorisation framework, the MediaPipe Holistic landmark extraction and sequence normalisation pipeline, the custom CNN dataset collection procedure, the LSTM model architecture and training, the real-time detection pipeline with majority voting smoothing, the per-attempt scoring and per-gesture evaluation logic for compound signs, the Flask REST API backend serving all inference and evaluation endpoints, the React frontend for vocabulary selection, dual-format sign demonstration, webcam capture, and feedback display, and the design and execution of the user acceptance testing protocol with hearing-impaired participants.

Samaraweera T.N. (IT22143754) designed and implemented the AI-Powered Sinhala Lip-Reading Assistant with 3D Avatar Feedback. This encompasses the controlled Sinhala lip-movement data collection protocol, the MediaPipe-based lip region ROI extraction and normalisation pipeline, the CNN-LSTM temporal sequence model for Sinhala lip-reading, the phoneme-to-viseme mapping framework, the GLB-based 3D avatar rendering and synchronisation system, the Sinhala-language feedback engine producing hint messages and stage-wise scores, the gamified progress dashboard with learning analytics, and the overall educational interaction design for visual speech practice.

Hettiarachchi D.S.W. (IT22255488) designed and implemented the Teacher-Student Engagement Module with Emotion-Adaptive Learning. This encompasses the DeepFace-based facial expression recognition system with the five-state emotional state machine, the HOG and facial landmark-based attention detection mechanism with automated video pause and resume functionality, the adaptive learning intervention engine with Sign Language quiz and visual matching exercise triggers, the NLP-based quiz generation pipeline including transcript generation, keyword extraction, and question generation, the Teacher Dashboard with real-time and aggregated emotional analytics and individual student attention reports, and the full-stack MERN implementation with Python Flask microservice integration.

Viduranga W.R.A.V. (IT22255488) designed and implemented the Resonance Multi-Sensory Music Experience Platform. This encompasses the CNN emotion recognition model trained on the DEAM dataset for continuous valence and arousal prediction from mel-spectrogram representations, the custom ESP32-based dual-motor haptic feedback device with PWM control, the browser-native haptic integration via the Web Serial API with beat-detection-driven motor control, the Google Cloud Speech-to-Text multi-language transcription subsystem with Sinhala Unicode validation, the dynamic emotion-reactive visual feedback system with HSL colour space interpolation, the Groq Llama 3.1 AI Music Guide chatbot with contextual injection, and the Playback Summary Report engine.

CHAPTER 4: CONCLUSION

4.1 Summary of the Research

This dissertation has presented the design, implementation, and evaluation of HearMe, a comprehensive, AI-powered inclusive e-learning platform for hearing-impaired students in Sri Lanka. The platform integrates four purpose-built modules — an Interactive SSL Self-Learning and Gesture Evaluation Module, an AI-Powered Sinhala Lip-Reading Assistant with 3D Avatar Feedback, a Teacher-Student Engagement Module with Emotion-Adaptive Learning, and the Resonance Multi-Sensory Music Experience Platform — within a unified, web-based architecture accessible without specialist hardware requirements.

The research addressed a critical and multidimensional gap in the existing landscape of educational technology for hearing-impaired students. By treating sign language learning, pronunciation guidance, emotional engagement monitoring, and music accessibility as interconnected and equally important dimensions of educational inclusion, the platform reflects the comprehensive nature of the barriers faced by hearing-impaired learners in mainstream digital educational environments.

Quantitative evaluation demonstrated SSL gesture classification accuracy between 88.89% and 100% across vocabulary categories, with majority voting smoothing achieving 94.1% real-time prediction consistency at 420 milliseconds latency. Emotion detection achieved approximately 80% classification accuracy with a 78% re-engagement rate following adaptive interventions. Music emotion recognition using the Dynamic Graph Rewiring Attention aggregator achieved the highest generalisation performance across all evaluation metrics. User acceptance testing with hearing-impaired participants produced consistently positive usability ratings, with the per-gesture feedback mechanism for compound SSL signs receiving the highest individual rating of 4.67 out of 5.

4.2 Research Contributions

The research makes the following specific contributions to the field of inclusive educational technology. First, the implementation of a dual-format SSL sign demonstration system combining real-world video with three-dimensional avatar rendering within a self-directed learning platform is novel and not present in any prior SSL learning tool identified in the literature review. Second, the hybrid LSTM and CNN pipeline with per-gesture evaluation logic for compound SSL signs represents a fine-grained feedback mechanism not previously implemented for SSL self-directed learning. Third, the integration of emotion detection, attention monitoring, and NLP-based adaptive Sign Language quiz generation within a unified, hardware-free web architecture addresses a gap identified across all prior systems in the literature. Fourth, the browser-native haptic hardware integration via the Web Serial API for real-time music accessibility eliminates the adoption barriers of prior haptic music systems. Fifth, the AI Music Guide's contextual injection pattern for music education provides a novel approach to context-aware AI-assisted accessible learning. Sixth, the unified platform architecture demonstrates that all four dimensions of educational exclusion can be addressed within a single, maintainable, modular web platform.

4.3 Recommendations for Future Work

Future development should prioritise vocabulary expansion for the SSL module to encompass the full SSL400 dataset of 400 sign classes, potentially employing transformer-based architectures for improved accuracy at scale. Mobile application deployment via Android and iOS would substantially expand accessibility for South Asian users where smartphone-first digital engagement is prevalent. Model personalisation through online learning and transfer learning from individual user data would improve recognition accuracy for individual users over time and address signer variability limitations identified in the evaluation.

Expanding the Sinhala lip-movement dataset and extending the phoneme-to-viseme mapping to cover a larger portion of the Sinhala phoneme inventory would increase the educational coverage and practical utility of the lip-reading assistant. Integration of a more detailed face avatar model would improve articulatory fidelity and instructional value. For the emotion-adaptive module, multi-modal emotion detection incorporating sign language inputs and expansion of the adaptive intervention content library would improve intervention quality and breadth.

Integration of gamification elements including spaced repetition scheduling, achievement systems, and adaptive difficulty selection would enhance the motivational and pedagogical effectiveness of the self-directed learning experience across all four modules. A longitudinal evaluation study tracking learning outcomes over an extended period with a representative sample of hearing-impaired students would provide stronger evidence of the platform's educational impact than the cross-sectional user acceptance testing conducted in this study.

4.4 Concluding Remarks

HearMe demonstrates that deep learning, computer vision, natural language processing, accessible hardware integration, and AI-powered conversational guidance can be combined within a unified, web-based platform to address the comprehensive educational exclusion of hearing-impaired students from key dimensions of the digital learning experience. The strong quantitative results and the consistently positive reception from hearing-impaired user testing participants confirm that the platform achieves its research objectives and addresses genuine and previously unmet needs.

The research contributes to the broader vision of equitable digital education for the hearing-impaired community in Sri Lanka and provides a replicable, extensible, and modular technical foundation for future development of inclusive e-learning systems. It is the sincere hope of the research team that this work contributes meaningfully to reducing the educational barriers faced by hearing-impaired students and advances the state of accessible technology in Sri Lanka and the wider South Asian region.

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GLOSSARY

Term	Definition
Adaptive Learning	An educational approach in which the content, difficulty, or format of learning materials is automatically adjusted in response to the learner's performance, engagement, or emotional state.
Affective Computing	A field of computer science concerned with the design of systems that can recognise, interpret, and simulate human emotions.
Avatar	A three-dimensional animated character used in the HearMe platform to demonstrate Sinhala lip movements and SSL signs from multiple angles and at adjustable speeds.
CNN	Convolutional Neural Network — a class of deep neural network primarily effective for processing image and spectrogram data through hierarchical spatial feature extraction.
DeepFace	An open-source Python library that provides a unified interface to multiple pre-trained deep facial analysis models for face detection, verification, and emotion recognition.
ESP32	A low-cost, low-power microcontroller developed by Espressif Systems used in the Resonance platform to control dual haptic vibration motors via PWM signals.
Haptic Feedback	Physical vibrotactile stimulation delivered through a device to convey information through the sense of touch, used in the Resonance platform to translate musical rhythm and energy to hearing-impaired users.
LSTM	Long Short-Term Memory — a type of recurrent neural network architecture capable of learning long-range temporal dependencies, used in the SSL gesture recognition and Sinhala lip-reading modules.
MediaPipe Holistic	A Google open-source framework for real-time, multi-person body pose, face, and hand landmark extraction from video streams, used for feature extraction in the SSL and lip-reading modules.
Mel-Spectrogram	A visual representation of audio signal frequency content over time, with frequency axis scaled according to the mel perceptual scale, used as input to the CNN music emotion recognition model.
MER	Music Emotion Recognition — a subfield of Music Information Retrieval concerned with computational prediction of the emotional content of music from audio features.
NLP	Natural Language Processing — a branch of artificial intelligence concerned with enabling computers to understand, interpret, and generate human language, used in the quiz generation pipeline.
Phoneme	The smallest unit of sound in a language that can distinguish meaning, used in the lip-reading module to define the granularity of pronunciation guidance.
SSL	Sri Lankan Sign Language — the primary visual-gestural language used by the hearing-impaired community in Sri Lanka, distinct from ASL and BSL.
Valence/Arousal	The two dimensions of Russell's Circumplex Model of Affect: Valence represents emotional pleasantness (positive to negative), while Arousal represents emotional energy (calm to excited).

Term	Definition
Viseme	A visual representation of a phoneme as observed in the shape and movement of the lips and mouth, used to construct the phoneme-to-viseme mapping in the lip-reading assistant.
Web Serial API	A browser API that enables web applications to communicate directly with serial devices over USB without native drivers, used in the Resonance platform for browser-to-ESP32 haptic device communication.

APPENDICES

Appendix A: System Feature List

Feature ID	Description	Module	Status
F001	SSL vocabulary word search and selection interface	SSL Module	Implemented
F002	Reference video player for selected SSL sign	SSL Module	Implemented
F003	3D avatar rendering of selected SSL sign	SSL Module	Implemented
F004	Webcam capture and live video display for gesture practice	SSL Module	Implemented
F005	MediaPipe Holistic landmark extraction in browser	SSL Module	Implemented
F006	30-frame sliding sequence buffer with zero-padding	SSL Module	Implemented
F007	Majority voting smoothing with window of 10 predictions	SSL Module	Implemented
F008	Per-attempt scoring with decreasing credit	SSL Module	Implemented
F009	Per-gesture feedback for Category 2 compound signs	SSL Module	Implemented
F010	Sinhala lip-reading vocabulary input and target display	Lip-Reading	Implemented
F011	CNN-LSTM lip recognition model inference	Lip-Reading	Implemented
F012	Phoneme-to-viseme feedback mapping engine	Lip-Reading	Implemented
F013	3D avatar lip demonstration with adjustable speed	Lip-Reading	Implemented
F014	Gamified progress dashboard with stage scoring	Lip-Reading	Implemented
F015	Real-time DeepFace emotion classification (5 states)	Emotion Module	Implemented
F016	Attention detection with video pause and resume	Emotion Module	Implemented
F017	Adaptive SSL quiz intervention on disengagement	Emotion Module	Implemented
F018	Visual matching exercise intervention on confusion	Emotion Module	Implemented
F019	NLP-based quiz generation from video transcripts	Emotion Module	Implemented

Feature ID	Description	Module	Status
F020	Teacher Dashboard with real-time emotional analytics	Emotion Module	Implemented
F021	CNN music emotion recognition (valence/arousal)	Resonance	Implemented
F022	Real-time multi-language lyrics transcription (11 languages)	Resonance	Implemented
F023	Custom ESP32 haptic device with dual PWM motors	Resonance	Implemented
F024	Browser-native haptic control via Web Serial API	Resonance	Implemented
F025	Dynamic HSL emotion-reactive visual background	Resonance	Implemented
F026	Groq Llama 3.1 AI Music Guide chatbot	Resonance	Implemented
F027	Playback Summary Report engine	Resonance	Implemented
F028	Shared user authentication and session management	Platform	Implemented
F029	Cross-module progress tracking in MongoDB	Platform	Implemented
F030	Mobile responsive layout	Platform	Planned

Appendix B: Emotion Detection State Machine

The emotion state machine governs transitions between emotional states in the Emotion-Adaptive Module. A state transition requires sustained detection of a new emotion over a configurable number of consecutive frames (default: 5 frames at approximately 0.5 seconds) before the system registers a state change. This prevents spurious transitions caused by momentary facial expressions unrelated to the learner's actual emotional state. Once a state transition is registered, the intervention timer begins. If the Bored or Disengaged state persists for more than a configurable threshold (default: 90 seconds), the adaptive intervention engine is triggered. The Happy and Neutral states are treated as baseline engagement states and do not trigger interventions. The Confused state triggers a gentler intervention after a longer sustained duration (default: 120 seconds) to distinguish genuine confusion from transient cognitive effort.

Appendix C: Sample User Acceptance Testing Questionnaire

The following questionnaire was administered to hearing-impaired participants following user testing sessions with the SSL Self-Learning and Gesture Evaluation Module. Responses were recorded on a 5-point Likert scale from 1 (Strongly Disagree) to 5 (Strongly Agree).

Question ID	Question	Scale
Q1	The interface was easy to navigate without external guidance.	1-5
Q2	The reference video clearly showed the correct way to perform each sign.	1-5
Q3	The 3D avatar helped me understand the hand shape from different angles.	1-5
Q4	The system's gesture evaluation feedback was clear and understandable.	1-5
Q5	For compound signs, identifying which specific gesture was incorrect was helpful.	1-5
Q6	The scoring system motivated me to practise signs until I performed them correctly.	1-5
Q7	The system's real-time response felt fast enough for comfortable practice.	1-5
Q8	I would use this system regularly to practise SSL vocabulary.	1-5
Q9	Overall, I am satisfied with the HearMe sign language learning module.	1-5
Q10	I feel this system would be useful for other hearing-impaired students.	1-5

Appendix D: Ethical Considerations and Data Privacy

All user testing conducted for the HearMe platform adhered to the following ethical principles. Informed consent was obtained from all participants prior to their involvement in testing sessions. Participants were informed of the purpose of the research, the nature of the data collected (webcam images for gesture and emotion recognition, performance scores, questionnaire responses), and their right to withdraw from the study at any time without consequence. For participants under the age of 18, written consent was obtained from a parent or guardian in addition to the participant's own assent.

All user-generated data collected during testing sessions was stored securely on password-protected institutional storage systems and accessed only by the research team. No raw video data was retained beyond the testing session; only extracted landmark sequences and anonymised performance scores were stored for analysis. Participant identities are not disclosed in this dissertation, and results are reported in aggregate form. The system's browser-based landmark extraction architecture ensures that raw webcam video is processed locally and is not transmitted to any backend server, minimising the data privacy risk of the deployed system.